

Fig. 10: Error message

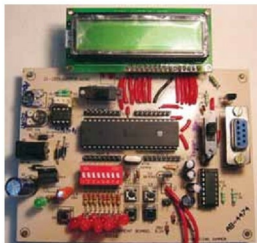


Fig. 11: Prototype of ATmega32 development board

chip PIC micro. For details, visit www.lancos.com/e2p/ponyprog2000.html

Example 1: Blinking LEDs

Include the following header files in the code to successfully compile the blinking-LED application program (blink_LED.c):

1. `<avr/io.h>` to include appropriate I/O definitions for the MCU
2. `<avr/interrupt.h>` to enable all the interrupt-handling functions (not used here).
3. `<util/delay.h>` to generate time delays

Also include a user-defined function (WaitMs()) to generate time delays in milliseconds. The parameter to be passed is an unsigned integer value indicating the time delay to be produced in milliseconds.

In the main function, first configure Port B pins as output pins. For this, set the corresponding bits in the DDRB register. After that, load 0xFF hex value to Port B register. This will make all the Port B pins to go high. Now generate a time delay of 500 ms using the WaitMs() function. Once the time delay is generated, all the Port B pins need to be cleared to switch off

the LEDs. For this, load 0x00 (complement of 0xFF) to Port B. The entire process repeats unconditionally inside a 'while' loop to generate the LED blinking effect.

Load your application program into the ATmega32 chip as follows:

1. Open AVR Studio. You will get a window as shown in Fig. 5. Click 'New Project' option.
2. Select AVR GCC and enter a name and destination folder for your project.
3. Select AVR simulator in 'Debug platform' window and ATmega32 in 'Device' window (refer Fig. 6).
4. Start writing your first program 'blink_LED.c' in the AVR Studio editor window. Configure the required options in 'Tools' and 'Project' menus.

5. Go to 'Build' menu and click 'Build' to compile the code. If there is no error, the hex code will generate.

6. Connect the ATmega32 development board to the serial port of the PC through the serial cable. On the board, set switch S7 to its 'program' position. Switch on the power supply. Run PonyProg and select 'Interface Setup' option from 'Setup' menu.

7. Set the I/O port as shown in Fig. 7.

Click 'Probe' button to check whether you have correctly connected the development board to the PC. A 'Test OK' message should appear. It will ensure that you have correctly connected the serial cable to the board, the power supply is turned on, switch S7 is turned to its 'program' position and you have chosen the correct COM port.

8. Select AVR Micro and ATmega32 options from 'Device Selection' menu in the toolbar (see Fig. 8).

9. Click 'Security and Configuration Bits.' Click 'Read' button to read the bits. Then make the settings as shown in Fig. 9. Now click 'Write.'

10. Open the hex file generated by AVR Studio. Go to 'Command' menu and click 'Write All' in PonyProg.

11. Click 'Yes' to start writing the code into the chip. Slide all the switches on the DIP-SW switch to 'On' position. If all the LEDs glow, you have completed the first application successfully. Otherwise, you will get an error message as shown in Fig. 10.

To solve the problem, ensure that the power is 'on,' switch S7 is in 'Program' mode and the serial cable is properly connected. Now check whether you have selected the correct device and the fuse-bit settings.

Fig. 11 shows the prototype of ATmega32 development board.

Example 2: Sequential light chaser effect

The source code (Chaser_Effect.c) for generating a sequential chaser effect on the LEDs (LED2 through LED9) is also included in the EFY-CD. It makes the LEDs connected to Port B of the MCU light up in a sequential manner back and forth.

Example 3: LCD interfacing

The LCD model described here is based on the HD44780 microcontroller from Hitachi and shows messages in two lines of 16 characters each. It displays all English alphabets, Greek letters, punctuation marks, mathematical symbols, etc. Every character consists of a 5x8 or 5x10 dot matrix. In our project, we have used a 5x8 character display, which is, in fact, the most commonly used.

Preset VR2 is used for contrast control. If there is no character display on the LCD, vary the preset to adjust the contrast. A 12-pin female header is used on the LCD module to connect the development board. On the board, a female header strip is used for LCD connection. Refer to the datasheet of the LCD module for different functions of the pins.

LCD memory. The LCD contains three memory blocks: DDRAM (display data RAM), CGROM (character